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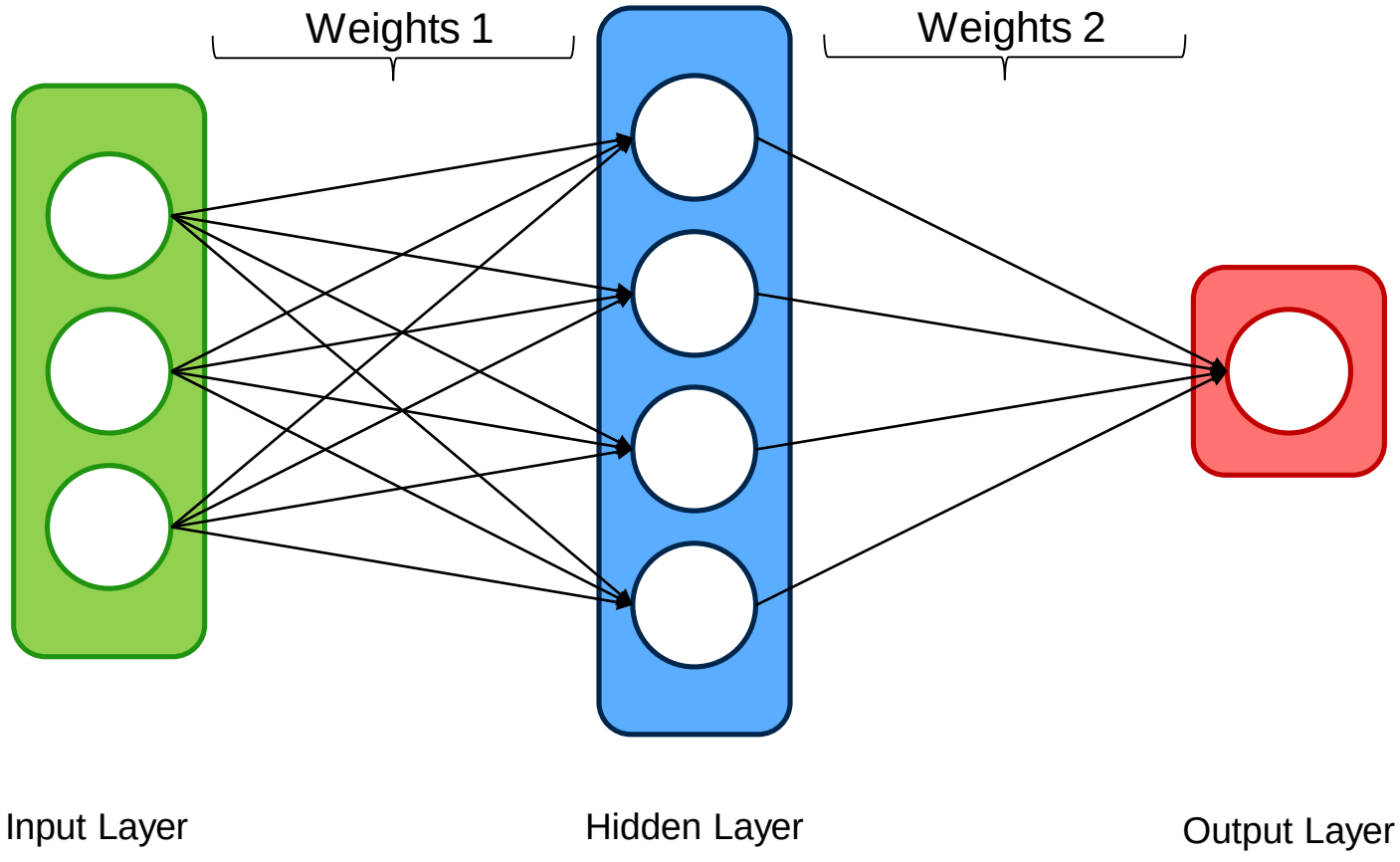
Deep Learning (for Computer Vision)

Arjun Jain

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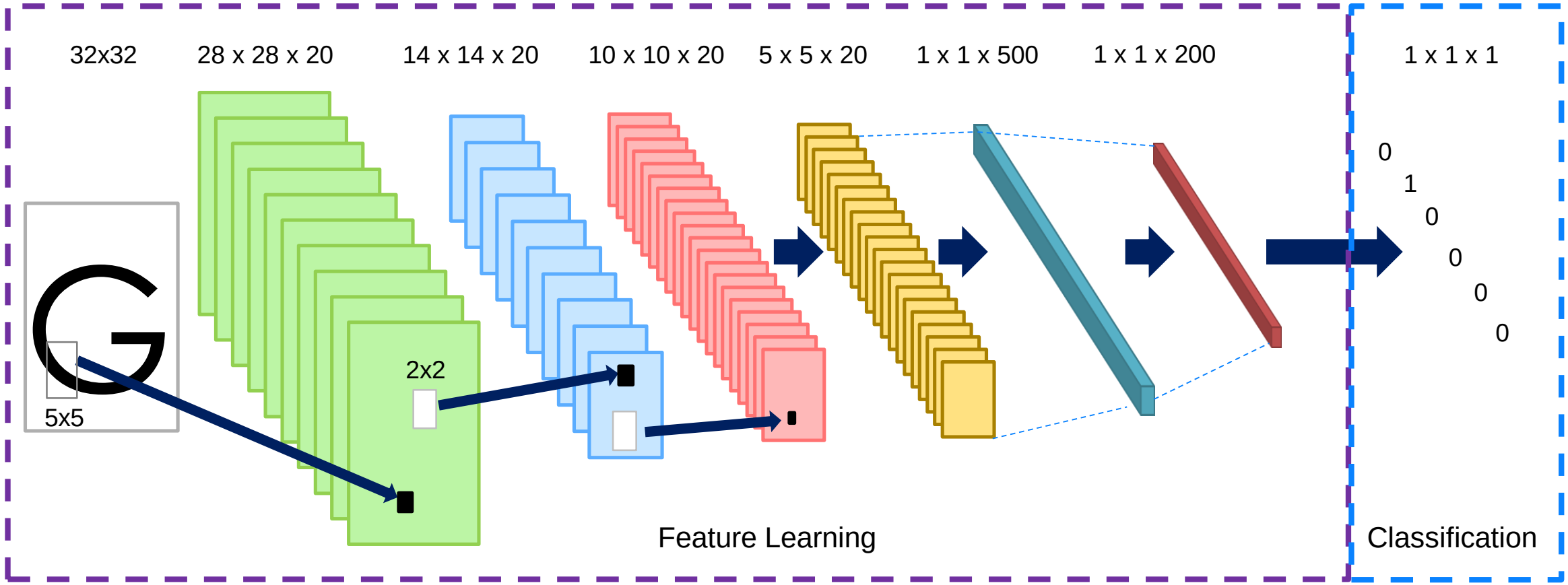
Convolution Network vs. Plain Neural Network

2 layered Plain Neural Network



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Convolution Network vs. Plain Neural Network



Input Layer | CONV layer | POOL layer | CONV layer | POOL layer | FC Layer | FC Layer | Output layer Fully Connected

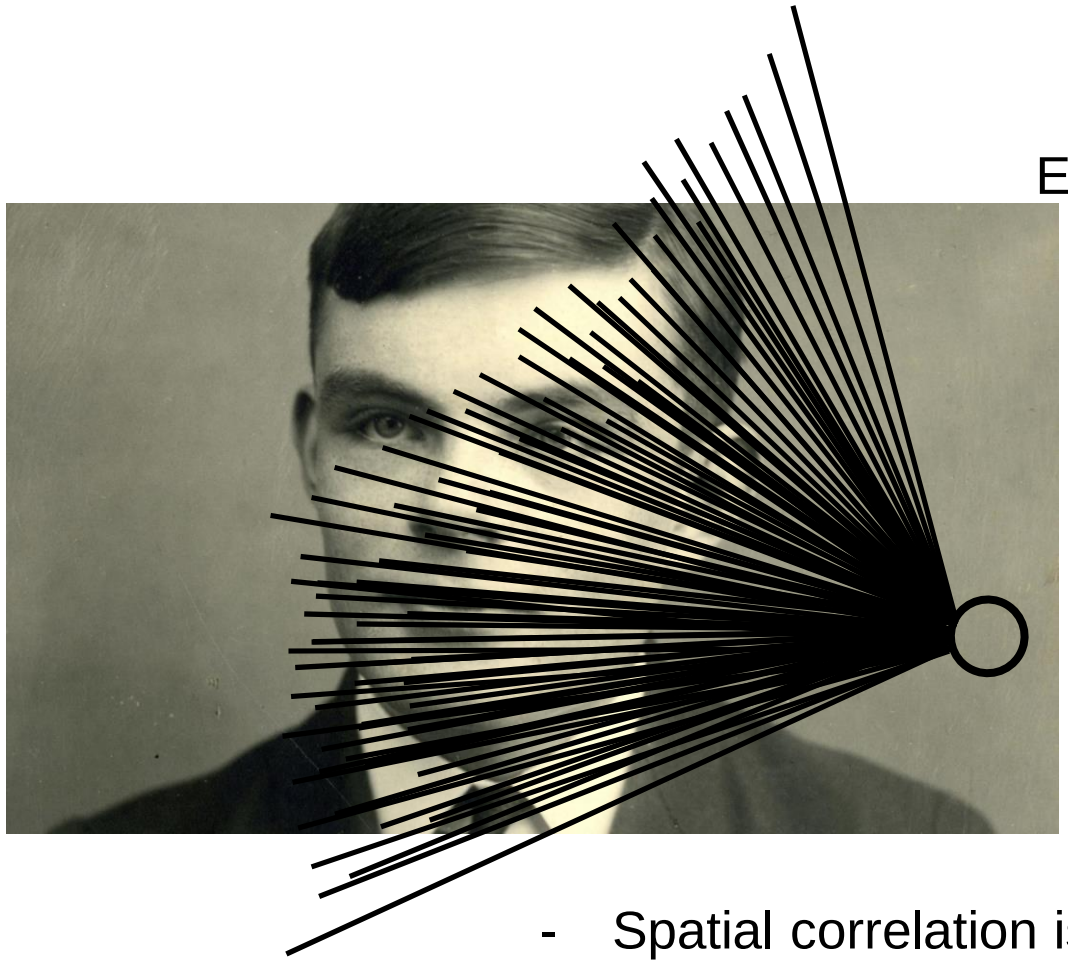
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Convolution

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Convolution



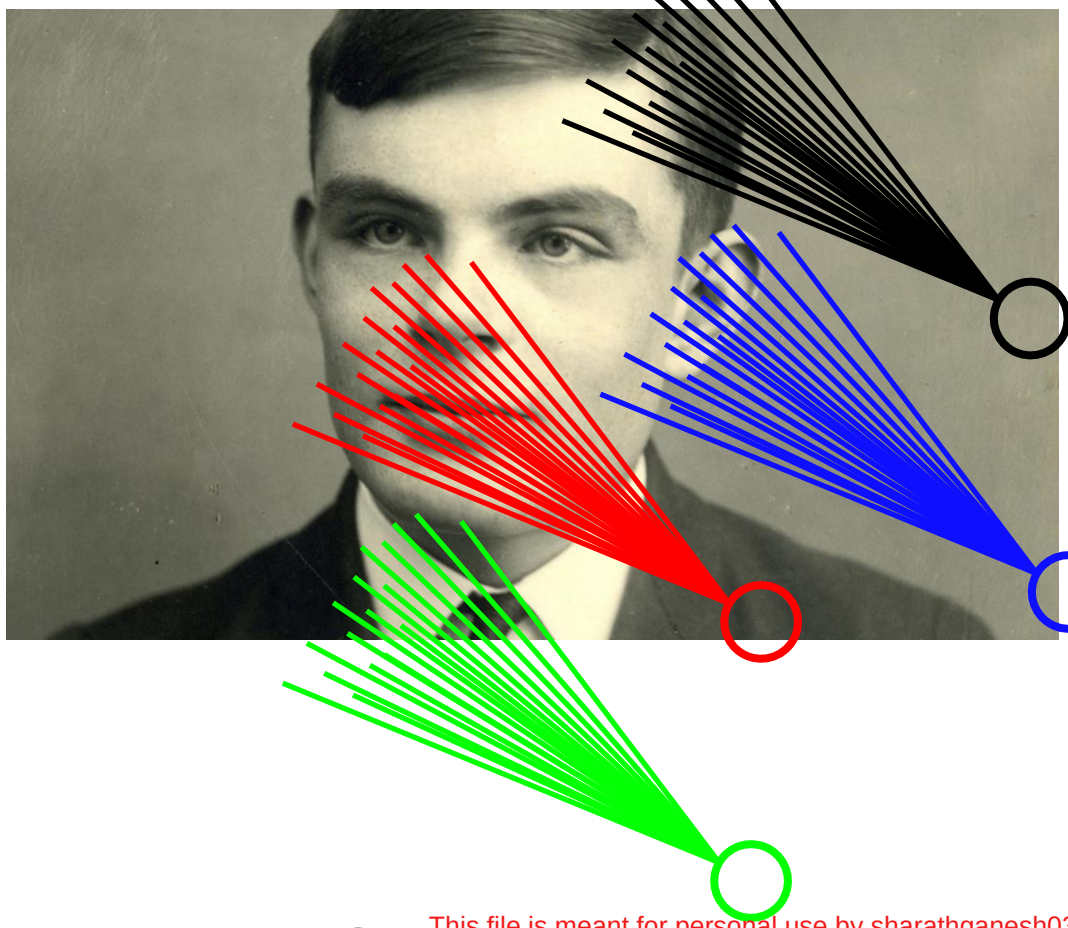
Example: 200x200 image
40K hidden units

➡ **~2B parameters!!!**

- Spatial correlation is local
- Waster of resources + we do not have enough training samples anyway

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Locally Connected Layer

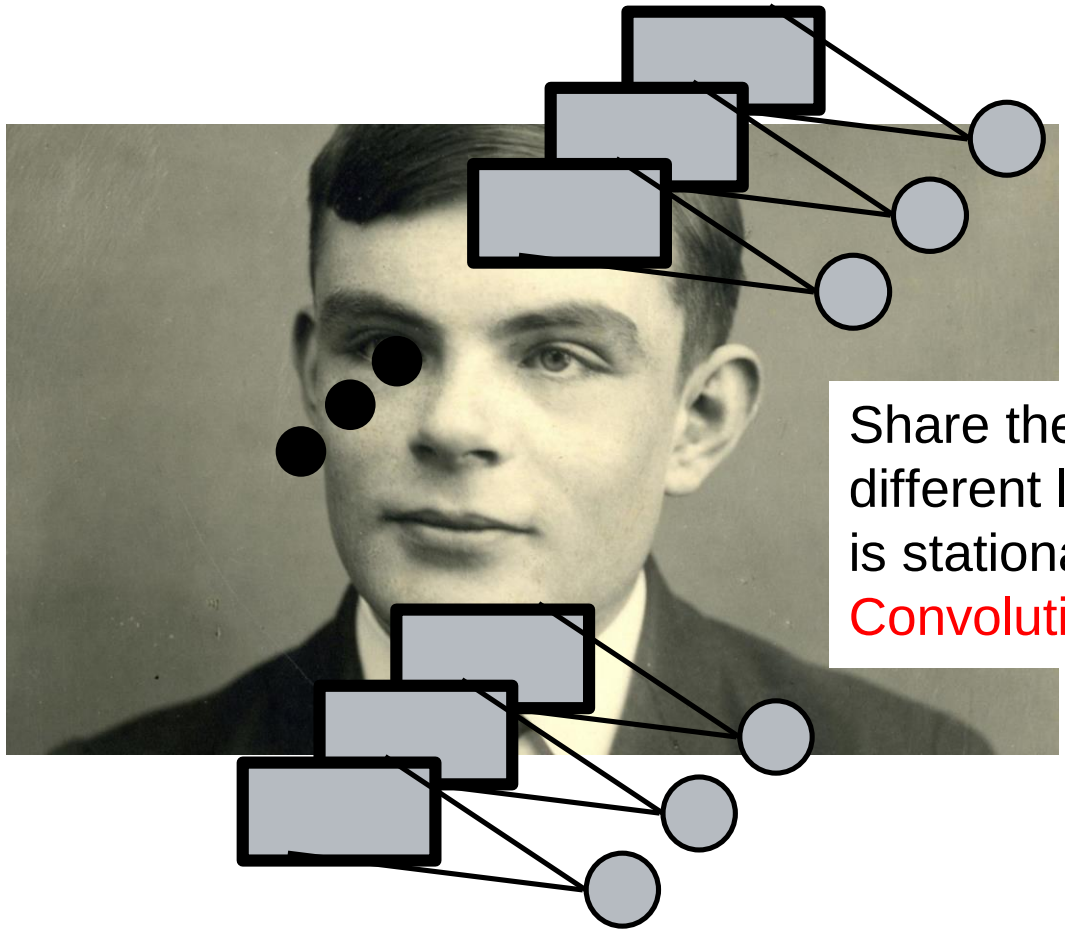


Example: 200x200 image
40K hidden units
Filter size: 10x10
4M parameters

Note: This parametrization is good when input image is registered (e.g. face recognition)

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Transitional Invariance

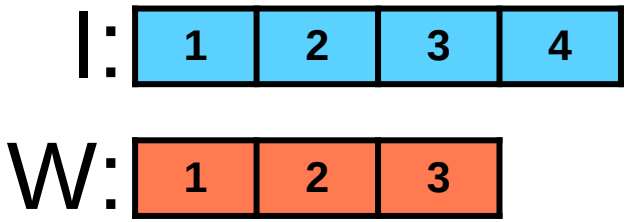


Share the same parameters across different locations (assuming input is stationary):

Convolutions with learned kernels

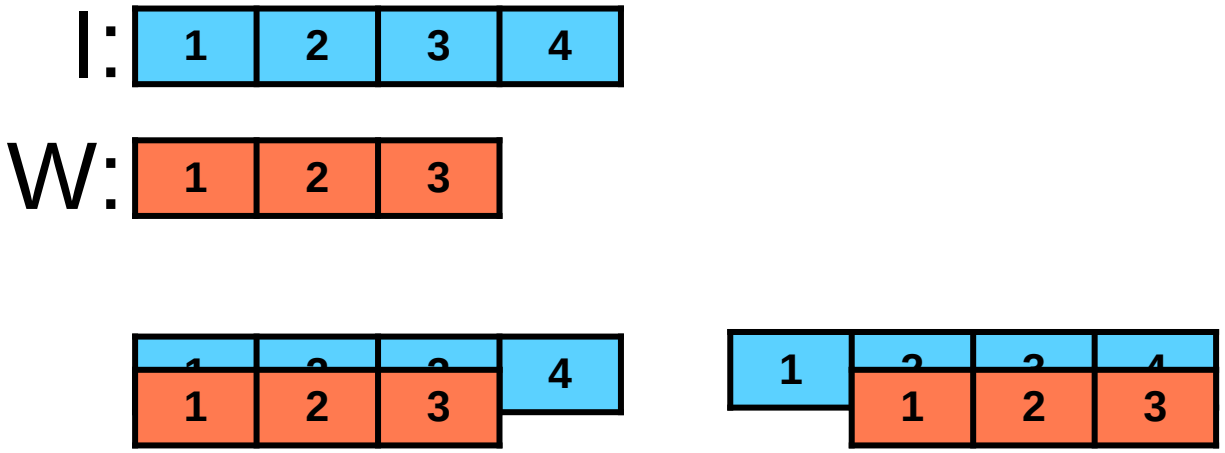
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Convolution (Discrete 1D)



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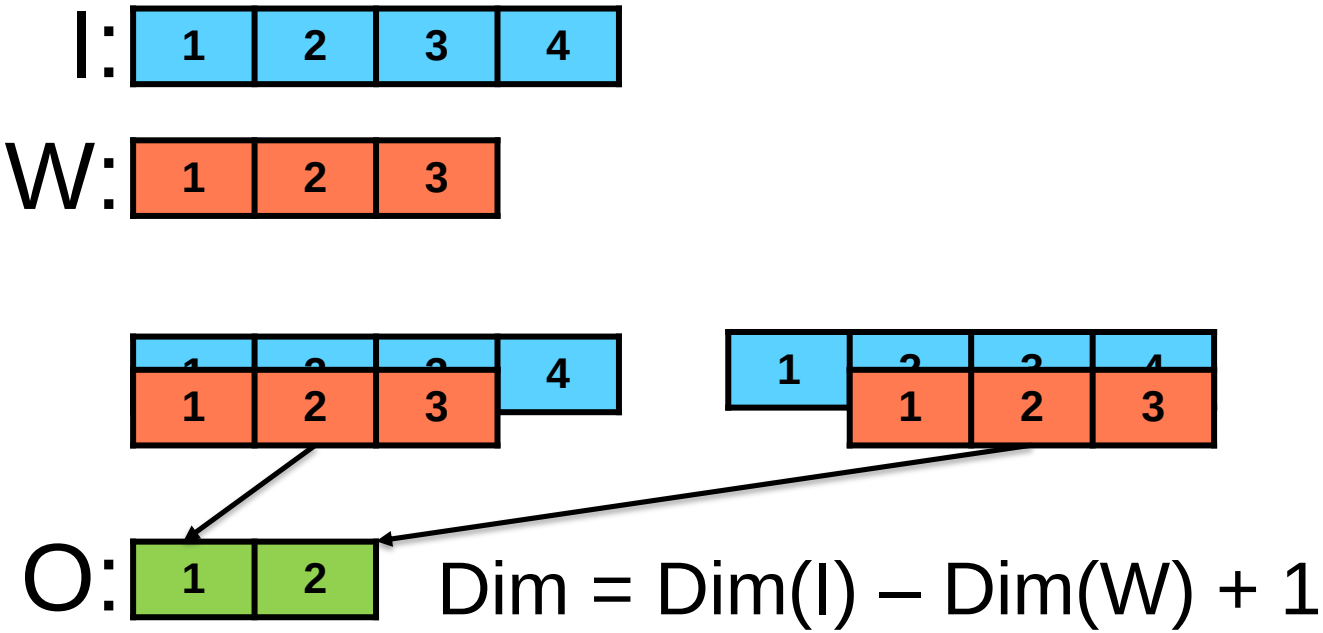
Convolution (Discrete 1D)



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Convolution (Discrete 1D)



Dim = Dim(I) – Dim(W) + 1

$$O_1 = I_1W_1 + I_2W_2 + I_3W_3$$

$$O_2 = I_2W_1 + I_3W_2 + I_4W_3$$

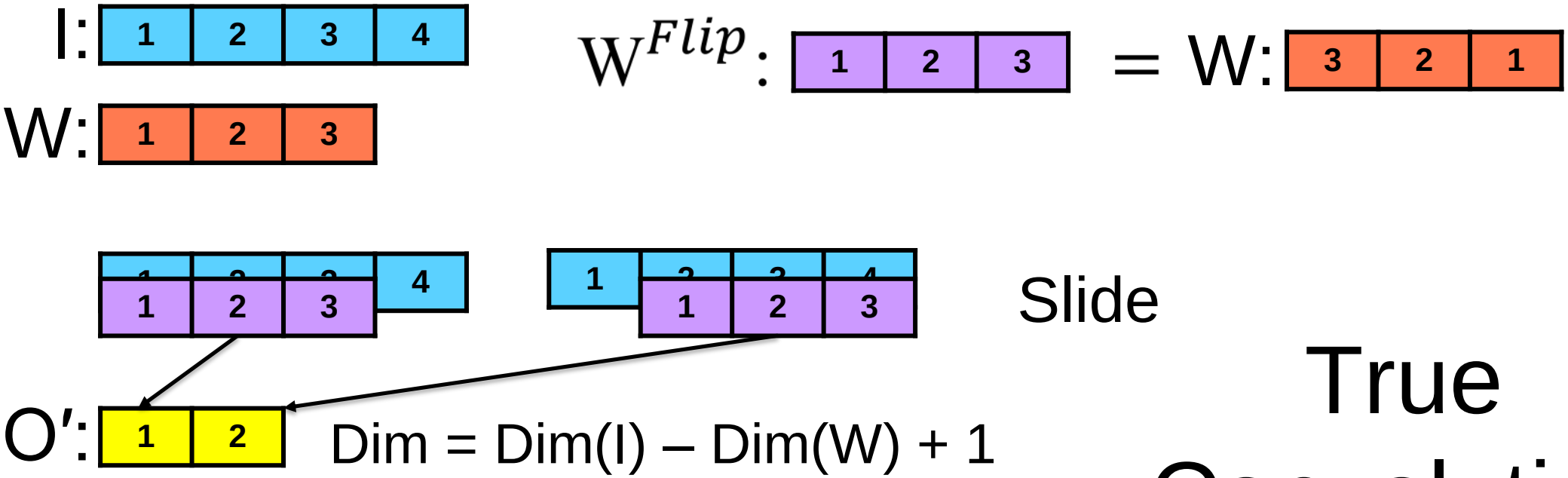
Correlation

$$O_i = \sum_{j=1}^{Dim(W)} I_{j+i-1}\, W_j$$

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Convolution (Discrete 1D)



Dim = Dim(I) – Dim(W) + 1

$$O'_1 = I_1 W_1^{Flip} + I_2 W_2^{Flip} + I_3 W_3^{Flip}$$

$$O'_2 = I_2 W_1^{Flip} + I_3 W_2^{Flip} + I_4 W_3^{Flip}$$

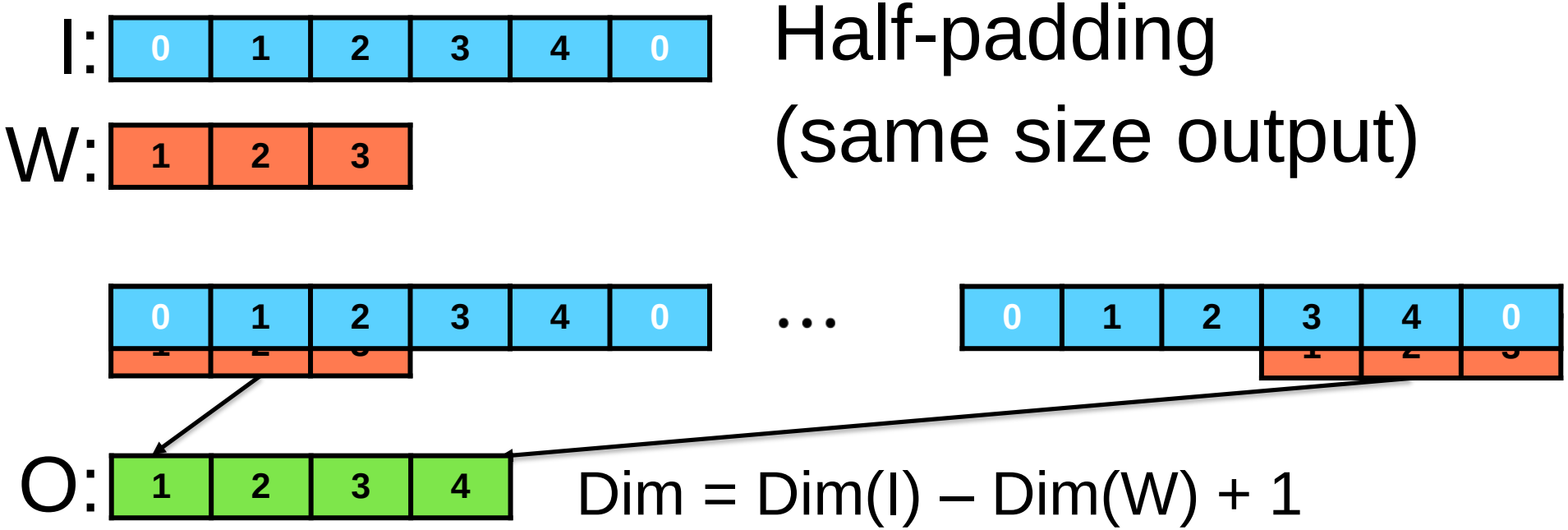
True Convolution

$$O_i = \sum_{j=1}^{Dim(W)} I_{j+i-1} W_{Dim(W)-j+1}$$

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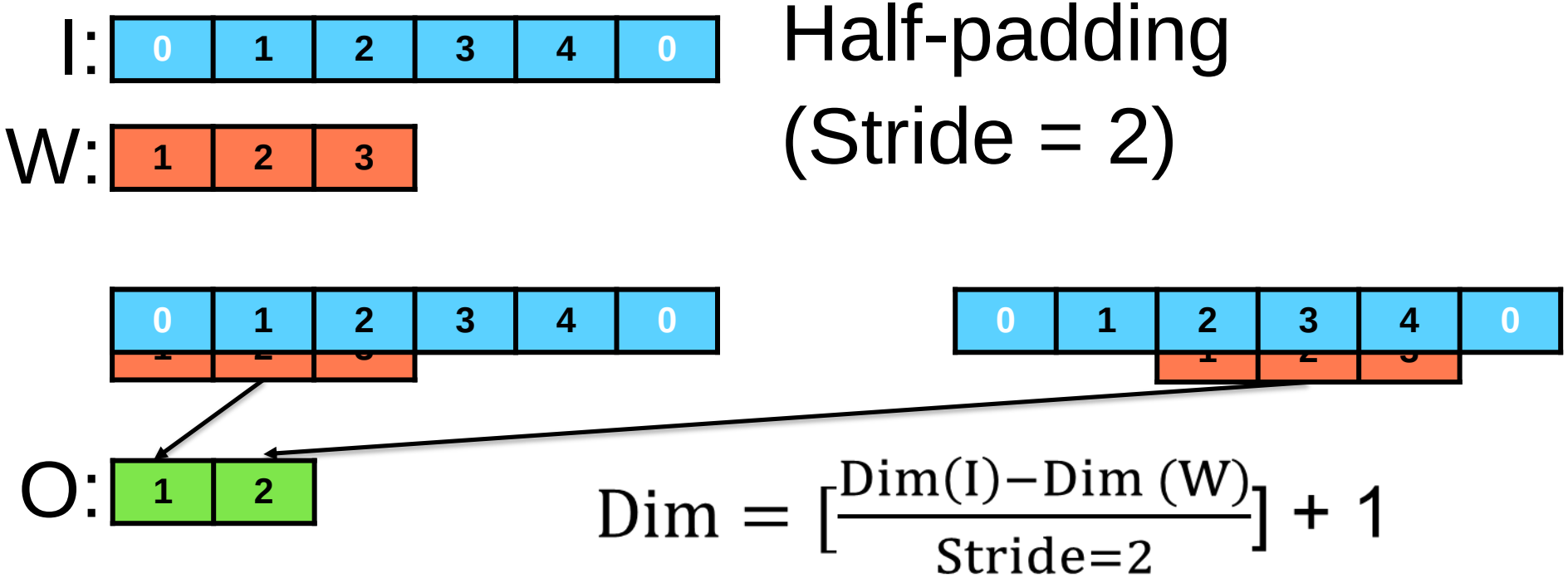
Convolution (Discrete 1D)



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Convolution (Discrete 1D)



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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

0.1
0.2

Bias **b** = 2
(nOutputPlane)

Image O = 2 x 3 x 3

O[1, :, :]

O[2, :, :]

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

2

1

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

3.1		

O[2, :, :]

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Spatial Dimension - Convolution

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(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-5

1

+

0.1

0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	

O[2, :, :]

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-10

-1

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9

O[2, :, :]

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-13

1

+

0.1

0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1		

O[2, :, :]

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-7

-3

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	

O[2, :, :]

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

14

2

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1

O[2, :, :]

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-4

0

+

0.1

0.2

Bias **b** = 2
(nOutputPlane)

Image O = 2 x 3 x 3

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9		

O[2, :, :]

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

1

-1

0.1

0.2

+

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	

O[2, :, :]

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

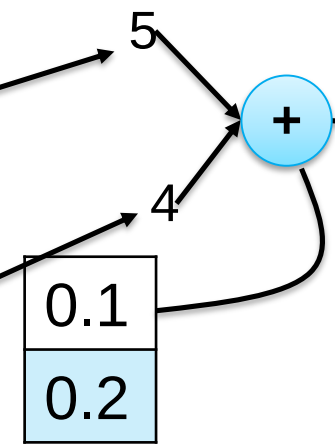
3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4



Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

7

-8

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8		

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

4

4

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

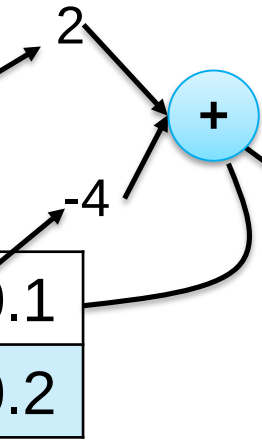
3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4



Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

9

-4

+

0.1

0.2

Bias **b** = 2
(nOutputPlane)

Image O = 2 x 3 x 3

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2
5.2		

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

6

12

+

Bias **b** = 2
(nOutputPlane)

0.1
0.2

Image O = 2 x 3 x 3

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2
5.2	18.2	

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

3

4

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2
5.2	18.2	7.2

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Spatial Dimension - Convolution

Image I = 2 x 4 x 4

Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-1

-8

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2
5.2	18.2	7.2
-8.8		

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

2

0

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

O[2, :, :]

-0.8	8.2	6.2
5.2	18.2	7.2
-8.8	2.2	

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Spatial Dimension - Convolution

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Weights W = 2 x 2 x 2 x 2
(nOutputPlane x nInputPlane x kH x kW)

Image O = 2 x 3 x 3

I[1, :, :]

1	-2	2	2
2	1	3	-2
-2	3	-3	1
-1	2	-4	2

I[2, :, :]

3	0	0	0
-2	-2	1	-1
2	-1	3	1
5	-2	0	1

W[1, 1, :, :]

1	-2
-2	1

W[2, 1, :, :]

3	1
2	2

W[1, 2, :, :]

1	0
0	1

W[2, 2, :, :]

0	0
0	4

-12

4

+

0.1
0.2

Bias **b** = 2
(nOutputPlane)

O[1, :, :]

-3.1	-3.9	-10.9
4.1	-12.9	16.1
-3.9	.1	9.1

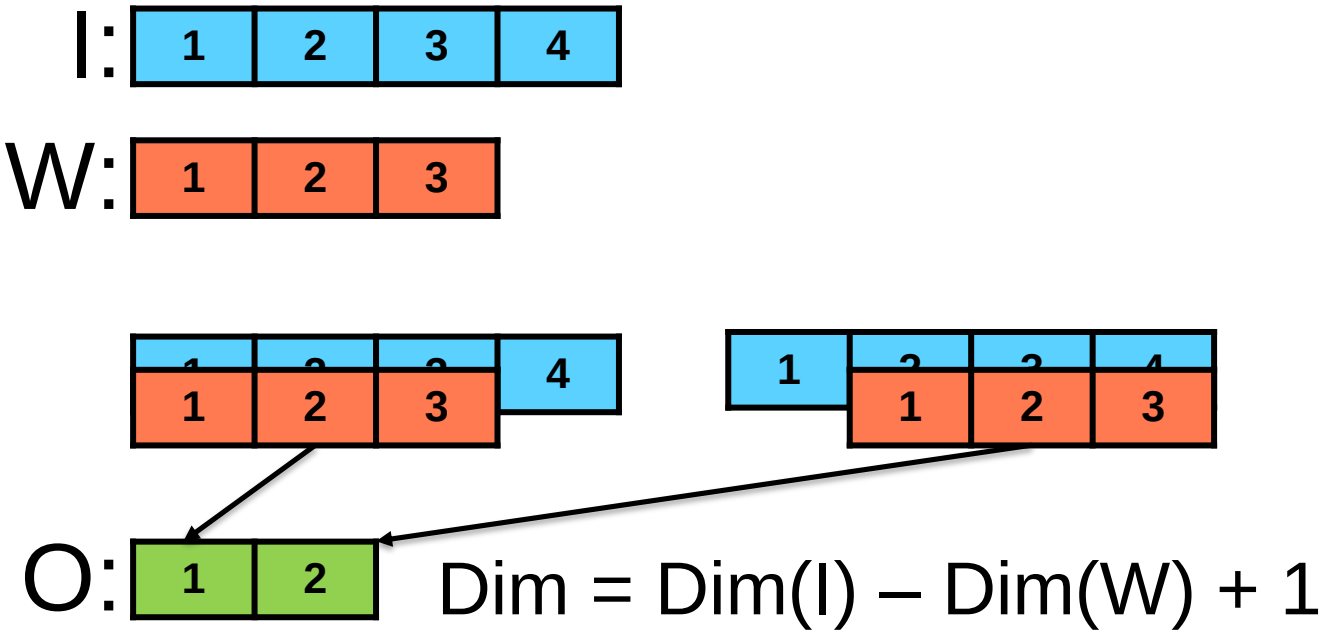
O[2, :, :]

-0.8	8.2	6.2
5.2	18.2	7.2
-8.8	2.2	-7.8

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Convolution - Backward



Dim = Dim(I) – Dim(W) + 1

$$O_1 = I_1 W_1 + I_2 W_2 + I_3 W_3$$

$$O_2 = I_2 W_1 + I_3 W_2 + I_4 W_3$$

Correlation

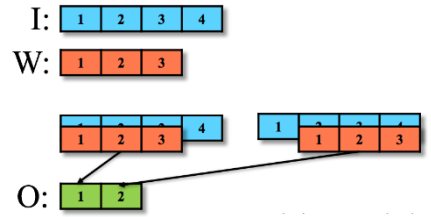
$$O_i = \sum_{j=1}^{Dim(W)} I_{j+i-1} W_j$$

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Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$



$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

Correlation

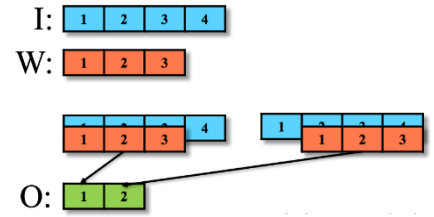
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Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$



$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

Correlation

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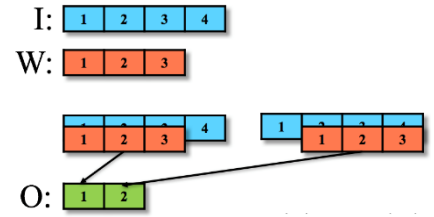
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Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial L O_1 \quad \partial L O_2]$$



$$O_1 = I_1 W_1 + I_2 W_2 + I_3 W_3$$

$$O_2 = I_2 W_1 + I_3 W_2 + I_4 W_3$$

Correlation

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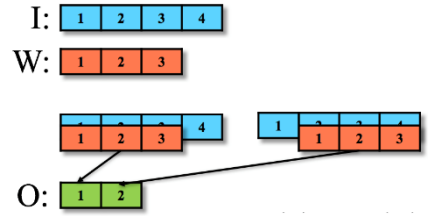
Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{W}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = [\partial LO_1 \times I_1 + \partial LO_2 \times I_2 \quad \partial LO_1 \times I_2 + \partial LO_2 \times I_3 \quad \partial LO_1 \times I_3 + \partial LO_2 \times I_4]$$



$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

Correlation

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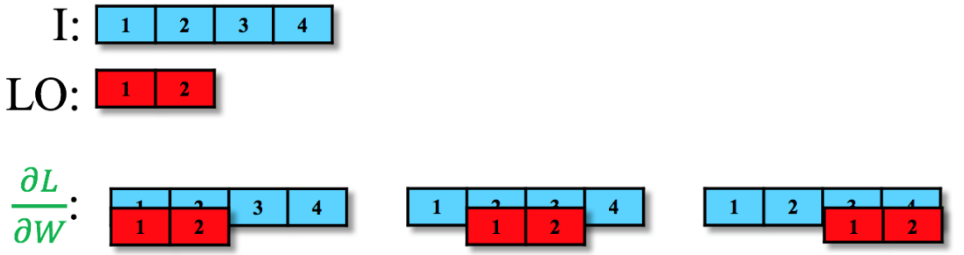
Convolution - Backward

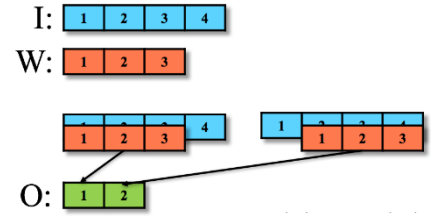
$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{W}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = [\partial LO_1 \times I_1 + \partial LO_2 \times I_2 \quad \partial LO_1 \times I_2 + \partial LO_2 \times I_3 \quad \partial LO_1 \times I_3 + \partial LO_2 \times I_4]$$





$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

Correlation

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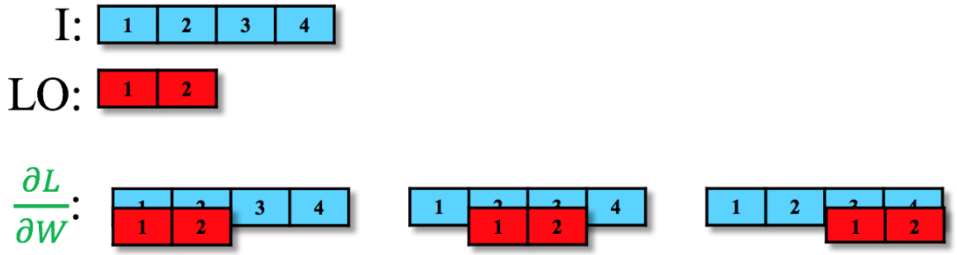
Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

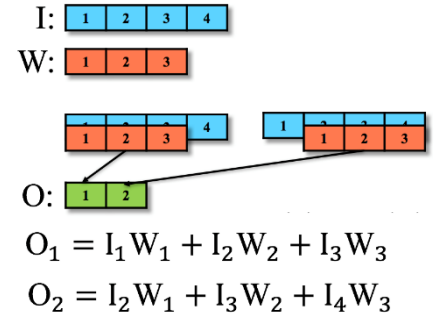
$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{W}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = [\partial LO_1 \times I_1 + \partial LO_2 \times I_2 \quad \partial LO_1 \times I_2 + \partial LO_2 \times I_3 \quad \partial LO_1 \times I_3 + \partial LO_2 \times I_4]$$



$$\frac{\partial L}{\partial \boldsymbol{W}} = \text{Correlation}(I, LO)$$



Correlation

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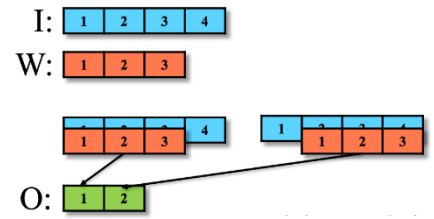
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Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial L O_1 \quad \partial L O_2]$$



$O_1 = I_1 W_1 + I_2 W_2 + I_3 W_3$

$O_2 = I_2 W_1 + I_3 W_2 + I_4 W_3$

Correlation

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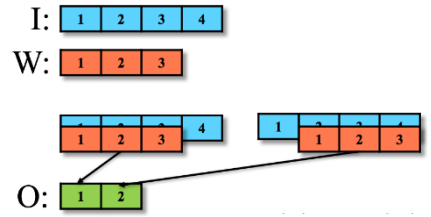
Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{I}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = [\partial LO_1 \times W_1 \quad \partial LO_1 \times W_2 + \partial LO_2 \times W_1 \quad \partial LO_1 \times W_3 + \partial LO_2 \times W_2 \quad \partial LO_2 \times W_3]$$



Correlation

$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

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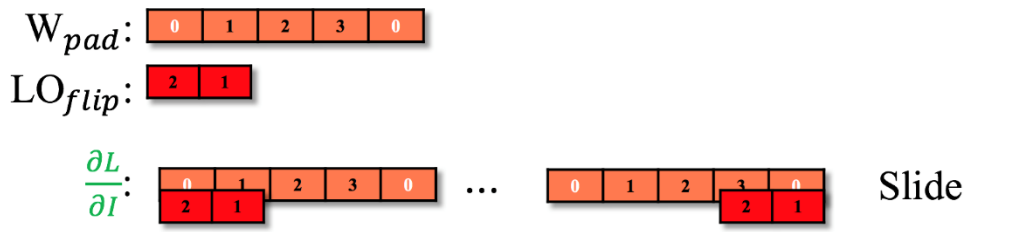
Convolution - Backward

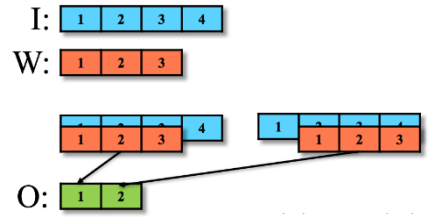
$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{W}} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_2 & I_3 & I_4 \end{bmatrix}$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{I}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = [\partial LO_1 \times W_1 \quad \partial LO_1 \times W_2 + \partial LO_2 \times W_1 \quad \partial LO_1 \times W_3 + \partial LO_2 \times W_2 \quad \partial LO_2 \times W_3]$$





$O_1 = I_1W_1 + I_2W_2 + I_3W_3$

$O_2 = I_2W_1 + I_3W_2 + I_4W_3$

Correlation

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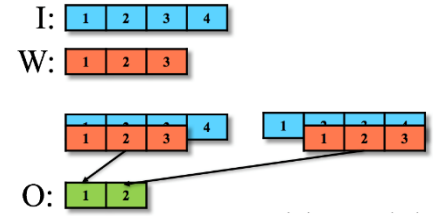
Convolution - Backward

$$\frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = \begin{bmatrix} W_1 & W_2 & W_3 & 0 \\ 0 & W_1 & W_2 & W_3 \end{bmatrix}$$

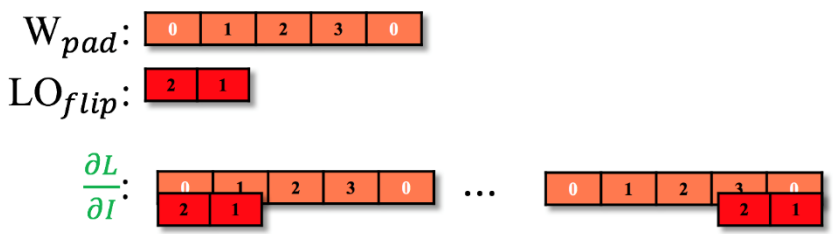
$$\frac{\partial L}{\partial \boldsymbol{I}} = \text{Correlation}(W_{pad}, \text{LO}_{flip})$$

$$\frac{\partial L}{\partial \boldsymbol{O}} = [\partial LO_1 \quad \partial LO_2]$$

$$\frac{\partial L}{\partial \boldsymbol{I}} = \frac{\partial L}{\partial \boldsymbol{O}} \times \frac{\partial \boldsymbol{O}}{\partial \boldsymbol{I}} = [\partial LO_1 \times W_1 \quad \partial LO_1 \times W_2 + \partial LO_2 \times W_1 \quad \partial LO_1 \times W_3 + \partial LO_2 \times W_2 \quad \partial LO_2 \times W_3]$$



Correlation



Slide

$$\frac{\partial L}{\partial \boldsymbol{I}} = \text{Correlation}(W_{pad}, \text{LO}_{flip})$$

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Convolution - Forward

```
n = tf.random_normal((1,4,1))
print(n)
```

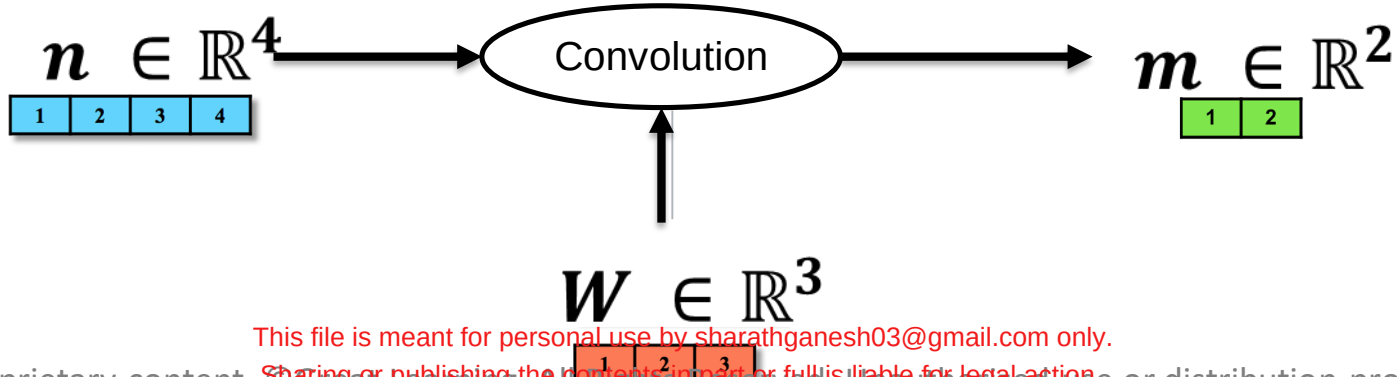
```
tf.Tensor(
[[[ 0.20350695]
  [-0.31481498]
  [-0.85656077]
  [ 0.5981919 ]]], shape=(1, 4, 1), dtype=float32)
```

```
nextgrad = tf.random_normal((1,2,1))
print(nextgrad)
```

```
tf.Tensor(
[[[-0.47567356]
  [-2.3547919 ]]], shape=(1, 2, 1), dtype=float32)
```

```
conv1 = tf.layers.Conv1D(1,3,1,'VALID',activation=None)
conv1.bias_initializer = tf.constant_initializer(0)
with tf.GradientTape(persistent=True) as t:
  t.watch(n)
  m = conv1(n)
  # A trick to make the gradOutput = nextgrad
  m1 = tf.multiply(nextgrad,m)
print(m)
```

```
tf.Tensor(
[[[-0.7662684]
  [ 0.2175143]]], shape=(1, 2, 1), dtype=float32)
```



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Convolution - Backward

```
#This is equivalent to conv:backward(n,nextgrad)
gradWeight = t.gradient(m1,conv1.weights[0])
print(gradWeight)
```

```
tf.Tensor(
[[[ 0.6445209]]

 [[ 2.1667714]]

 [[-1.0011742]]], shape=(3, 1, 1), dtype=float32)
```

```
conv1_back = tf.layers.Conv1D(1,2,1,'VALID',activation=None)
```

```
conv1_back.bias_initializer = tf.constant_initializer(0)
conv1_back.kernel_initializer = tf.constant_initializer(nextgrad.numpy())
gradWeight = conv1_back(n)
print(gradWeight)
```

```
tf.Tensor(
[[[ 0.6445209]
  [ 2.1667714]
  [-1.0011742]]], shape=(1, 3, 1), dtype=float32)
```

$$\frac{\partial L}{\partial m} \in \mathbb{R}^{1\times 2}$$

$$n \in \mathbb{R}^4$$



Convolution

$$m \in \mathbb{R}^2$$



$$W \in \mathbb{R}^3$$

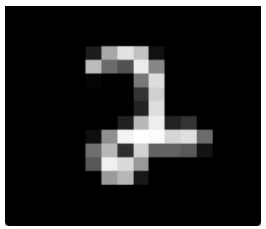


$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial m} \times \frac{\partial m}{\partial W}$$

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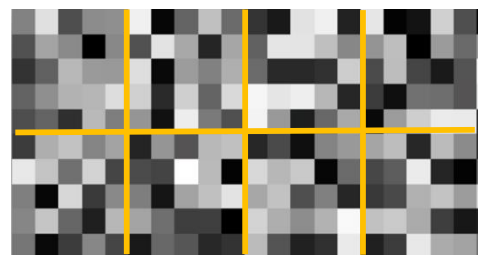
Convolution

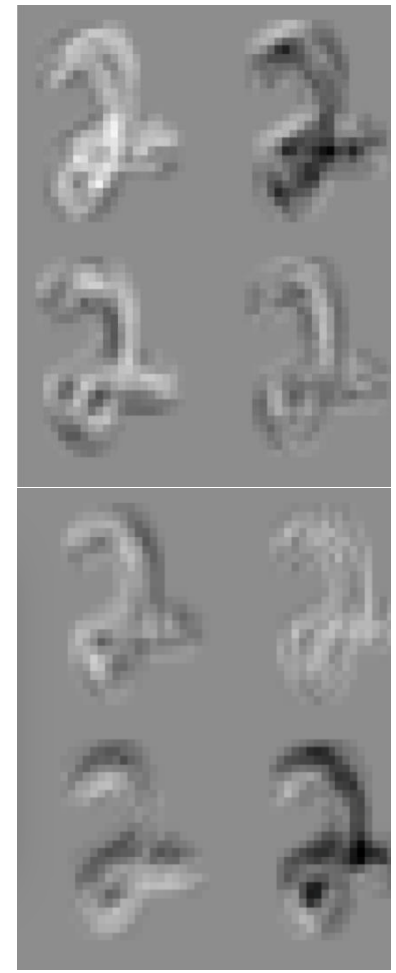


$$n \in \mathbb{R}^{16\times16} \rightarrow \text{Convolution} \rightarrow m \in \mathbb{R}^{8\times12\times12}$$

$$W \in \mathbb{R}^{1\times8\times5\times5}$$

$$=$$

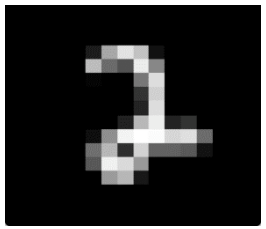




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Convolution



$$n \in \mathbb{R}^{16\times16} \rightarrow \text{Convolution} \rightarrow m \in \mathbb{R}^{8\times12\times12}$$

$$W$$

```
n = tf.random_normal((1,16,16,1))
print(n.shape)
```

```
(1, 16, 16, 1)
```

```
conv2 = tf.layers.Conv2D(8,[5,5],[1,1],
        "valid",activation=None)
```

```
m = conv2(n)
print(m.shape)
```

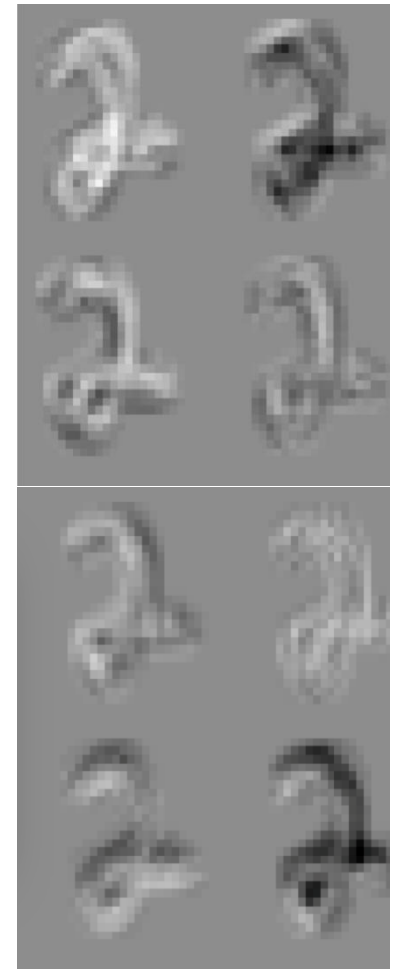
```
(1, 12, 12, 8)
```

```
print(conv2.weights[0].shape)
```

```
(5, 5, 1, 8)
```

```
print(conv2.weights[1].shape)
```

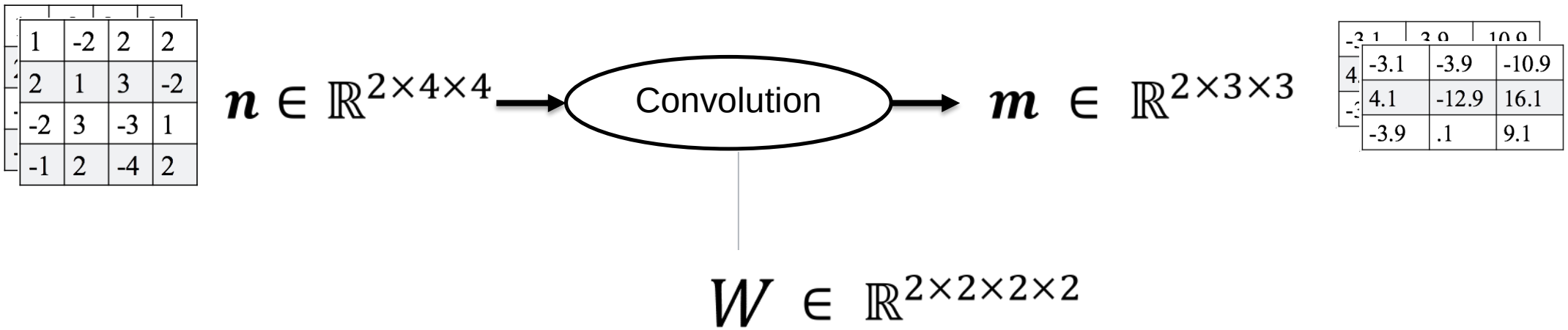
```
(8,)
```



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Convolution



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Convolution – in Tensorflow

```
I = tf.constant([[[[1,-2],[2,1],[-2,3],[-1,2]],[[2,2],[3,-2],[-3,1],[-4,2]],[[3,0],[-2,-2],[2,-1],[5,-2]],[[0,0],[1,-1],
                  [3,1],[0,1]]]], dtype=tf.float32)
```

```
print(I)
```

```
conv = tf.layers.Conv2D(2,[2,2],[1,1],"valid",activation=None)
```

```
conv.bias_initializer = tf.constant_initializer([0.1,0.2])
conv.kernel_initializer = tf.constant_initializer([[[[ 1, -2],[-2,  1]],[[1,  0],[0,  1]]],[[[3, 1],
                                  [2,  2]],[[0,  0],[ 0, 4]]]], dtype=tf.float32)
O = conv(I)
print(I.numpy()[0,:,:,:])
# conv.kernel_initializer = tf.constant_initializer()
```

```
[[[ 1. -2.]
  [ 2.  1.]
  [-2.  3.]
  [-1.  2.]]

 [[ 2.  2.]
  [ 3. -2.]
  [-3.  1.]
  [-4.  2.]]

 [[ 3.  0.]
  [-2. -2.]
  [ 2. -1.]
  [ 5. -2.]]]
```

```
print(O.numpy()[0,:,:,:])
```

```
[[[ 17.1  -4.8]
  [  3.1   3.2]
  [-15.9  16.2]]

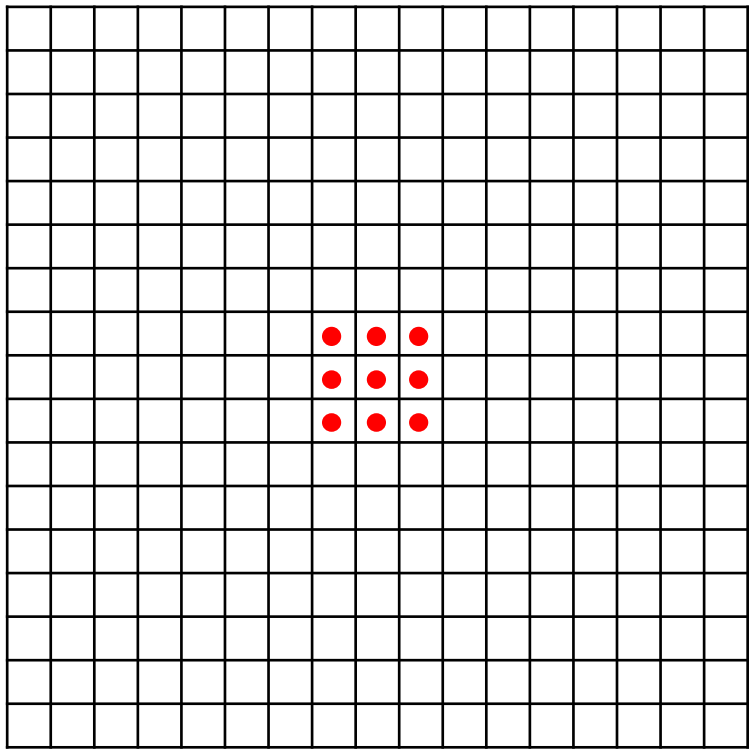
 [[ 10.1  -8.8]
  [ -5.9 -16.8]
  [ -4.9   1.2]]

 [[  1.1 -11.8]
  [  5.1   4.2]
  [ 20.1   2.2]]]
```

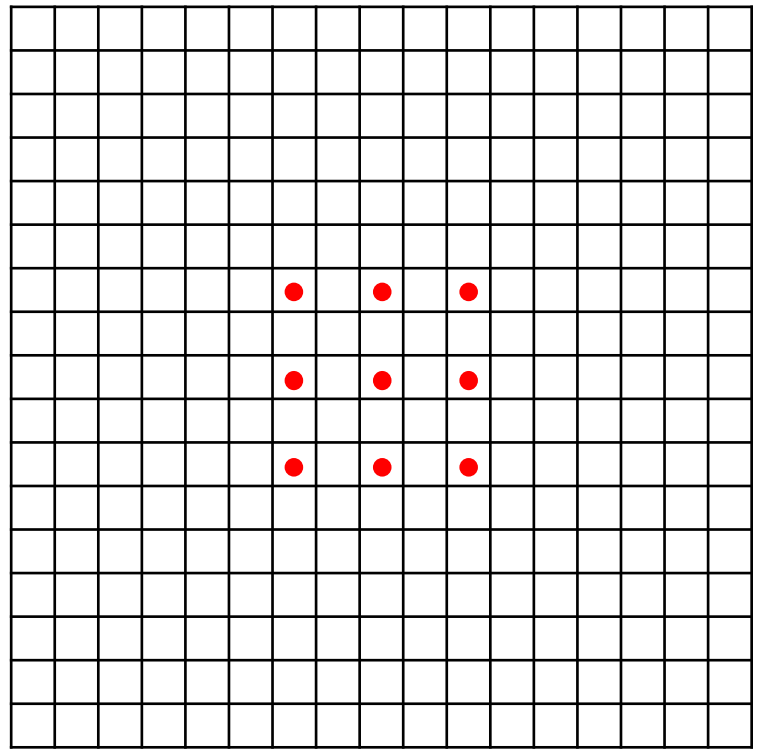
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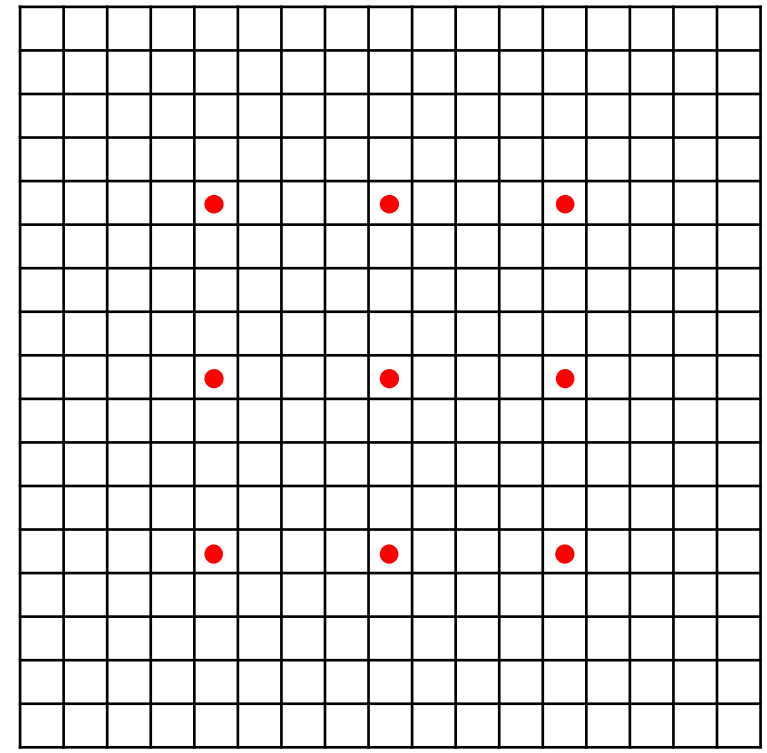
Aside: Dilated Convolution



(a)



(b)



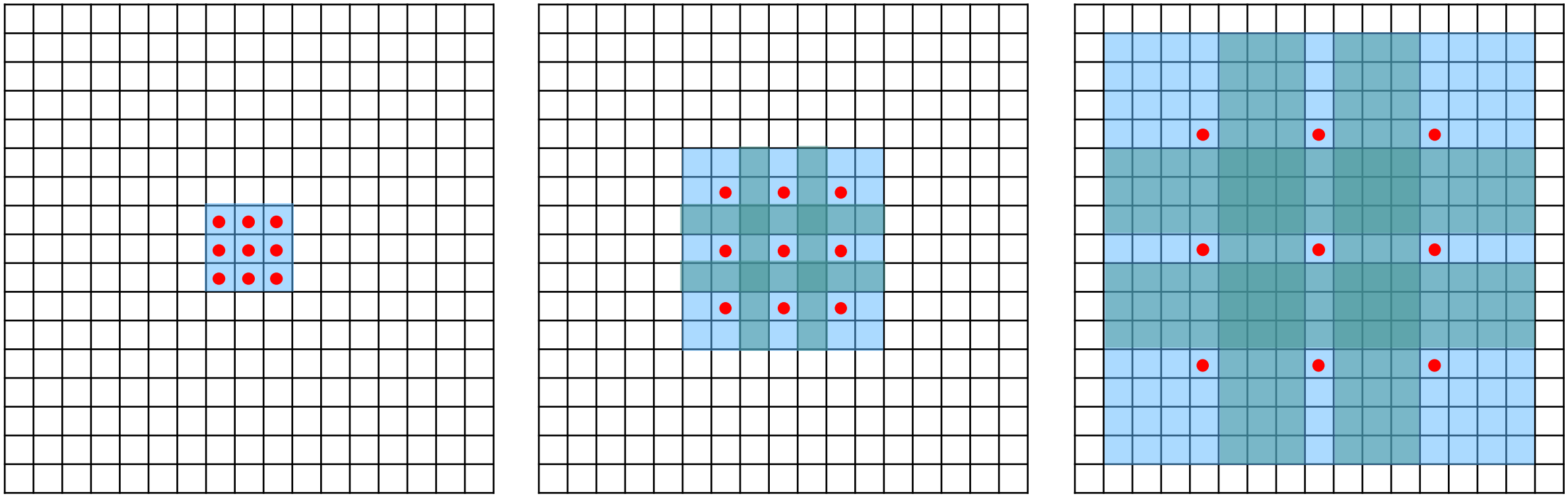
(c)

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Aside: Dilated Convolution

nn.SpatialDilatedConvolution



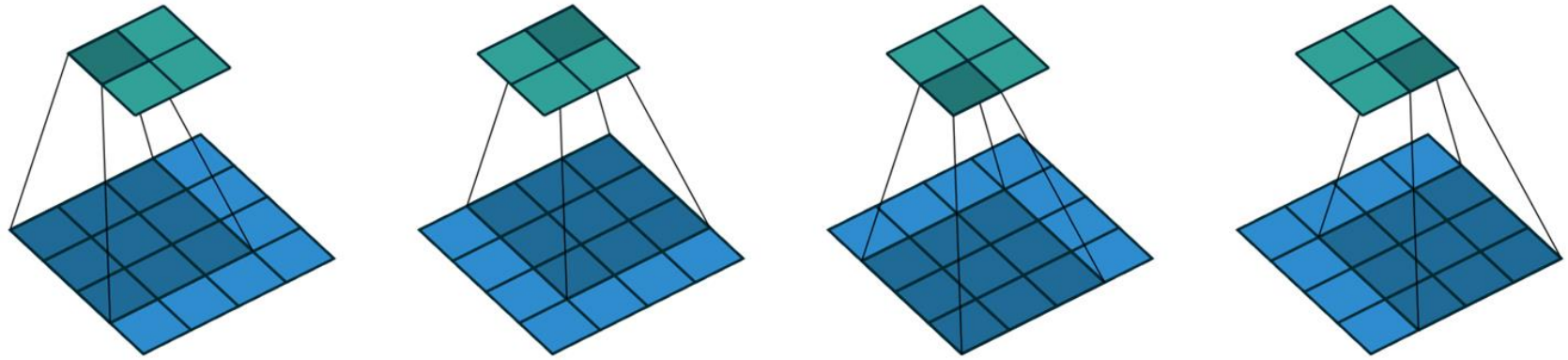
(a) (b) (c)

Source: Multi-Scale Context Aggregation by Dilated Convolutions Fisher Yu, Vladlen Koltun`

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Convolution



(No padding, unit strides) Convolving a 3 × 3 kernel over a 4 × 4 input using unit strides (i.e., i = 4, k = 3, s = 1 and p = 0).

$$\begin{pmatrix} w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 & 0 \\ 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} & 0 \\ 0 & 0 & 0 & 0 & 0 & w_{0,0} & w_{0,1} & w_{0,2} & 0 & w_{1,0} & w_{1,1} & w_{1,2} & 0 & w_{2,0} & w_{2,1} & w_{2,2} \end{pmatrix} \begin{bmatrix} I_0 \\ I_1 \\ \vdots \\ I_{15} \end{bmatrix}$$

$$\frac{\partial L}{\partial I} = W^T \cdot LO = Deconvolution$$

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Convolution

```
conv = tf.layers.Conv2D(1,[3,3],[2,2],"valid",activation=None)
w = tf.random_uniform((3,3),0,2)
conv.kernel_initializer = tf.constant_initializer(w.numpy())
conv.bias_initializer = tf.constant_initializer(0)
print(w)
```

```
tf.Tensor(
[[0.13429713 1.9649408  0.74939513]
 [0.6982393  1.2397809  1.4247327 ]
 [1.5534406  0.1190145  1.3256309 ]], shape=(3, 3), dtype=float32)
```

Normal Convolution Forward pass

```
imgC = tf.random_uniform((1,5,5,1),0,5)
with tf.GradientTape(persistent=True) as t:
  t.watch(imgC)
  r = conv(imgC)
  # Trick to give r as gradOutput
  r_1 = tf.multiply(0.5*r,r)
print(r)
```

```
tf.Tensor(
[[[[25.535583]
   [28.583742]]

  [[23.470192]
   [20.622599]]]], shape=(1, 2, 2, 1), dtype=float32)
```

Normal Convolution Backward pass

```
# This is equivalent to conv:backward(imgC,r)
grad_input = t.gradient(r_1,imgC)
print(grad_input)
```

```
tf.Tensor(
[[[[ 3.4293556]     [[42.819992 ]
   [50.17591  ]      [49.156643 ]
   [22.974957 ]      [98.61191  ]
   [56.16536  ]      [43.924065 ]
   [21.420517 ]]     [53.345966 ]]

  [[17.829948 ]     [[16.387812 ]
   [31.65853  ]      [29.097897 ]
   [56.339676 ]      [47.83826  ]
   [35.437576 ]      [25.567505 ]
   [40.724194 ]]     [29.381691 ]]

                    [[36.45955  ]
                     [ 2.7932932]
                     [63.148792 ]
                     [ 2.4543884]
                     [27.337954 ]]]], shape=(1, 5, 5, 1), dtype=float32)
```

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Convolution

```
trancon = tf.layers.Conv2DTranspose(1,[3,3],[2,2],"valid",activation=None)
trancon.bias_initializer = tf.constant_initializer(0)
trancon.kernel_initializer = tf.constant_initializer(w.numpy())

imgC_1 = trancon(r)
print(imgC_1)
```

```
tf.Tensor(
[[[[ 3.4293556]    [[42.819992 ]
   [50.17591  ]     [49.156643 ]
   [22.974957 ]     [98.61191  ]
   [56.16536  ]     [43.924065 ]
   [21.420517 ]]    [53.345966 ]]

  [[17.829948 ]    [[16.387812 ]
   [31.65853  ]     [29.097897 ]
   [56.339676 ]     [47.83826  ]
   [35.437576 ]     [25.567505 ]
   [40.724194 ]]    [29.381691 ]]

                    [[36.45955  ]
                     [ 2.7932932]
                     [63.148792 ]
                     [ 2.4543884]
                     [27.337954 ]]]],

shape=(1, 5, 5, 1), dtype=float32)
```

Forward pass is same as backward pass of normal convolution

Normal Convolution Backward pass

```
# This is equivalent to conv:backward(imgC,r)
grad_input = t.gradient(r_1,imgC)
print(grad_input)
```

```
tf.Tensor(
[[[[ 3.4293556]    [[42.819992 ]
   [50.17591  ]     [49.156643 ]
   [22.974957 ]     [98.61191  ]
   [56.16536  ]     [43.924065 ]
   [21.420517 ]]    [53.345966 ]]

  [[17.829948 ]    [[16.387812 ]
   [31.65853  ]     [29.097897 ]
   [56.339676 ]     [47.83826  ]
   [35.437576 ]     [25.567505 ]
   [40.724194 ]]    [29.381691 ]]

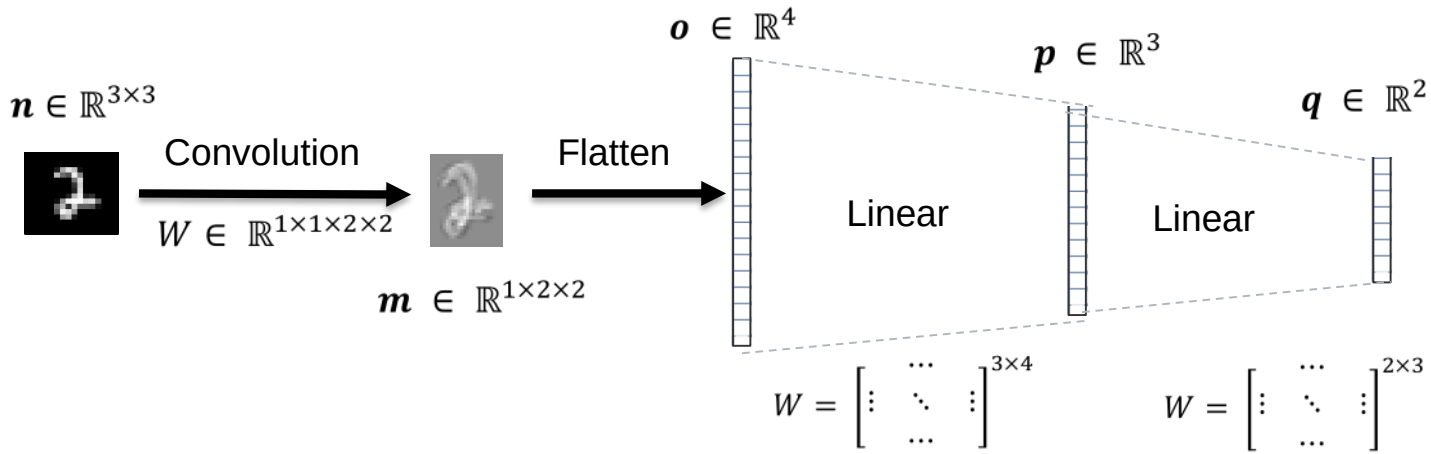
                    [[36.45955  ]
                     [ 2.7932932]
                     [63.148792 ]
                     [ 2.4543884]
                     [27.337954 ]]]],

shape=(1, 5, 5, 1), dtype=float32)
```

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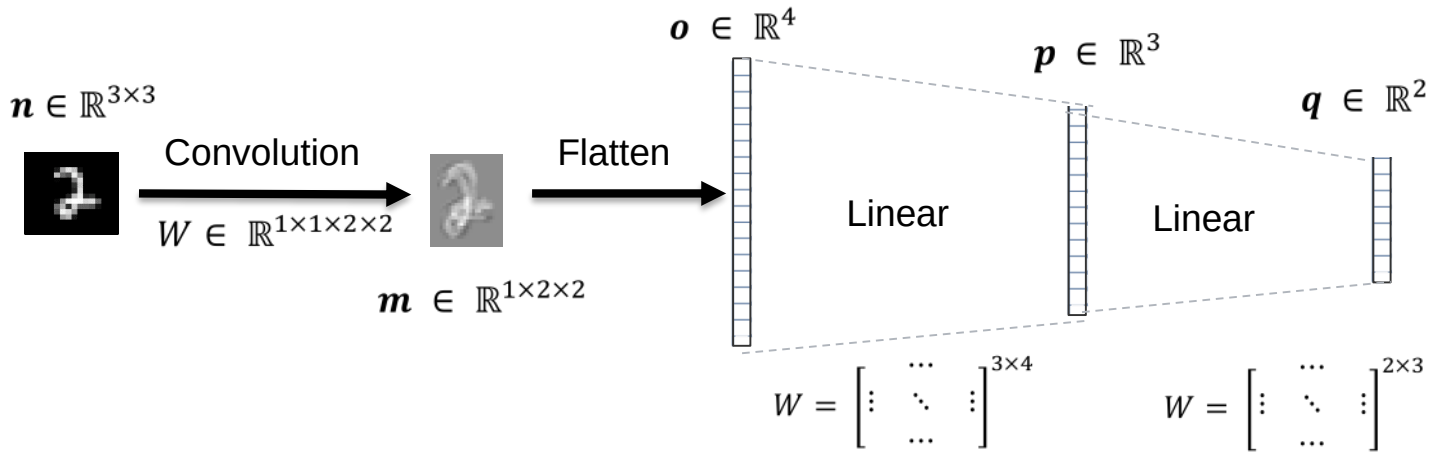
Everything is a Convolution!

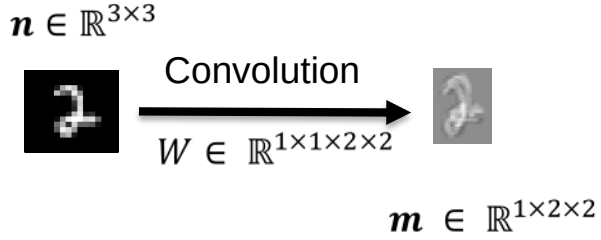


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Everything is a Convolution!

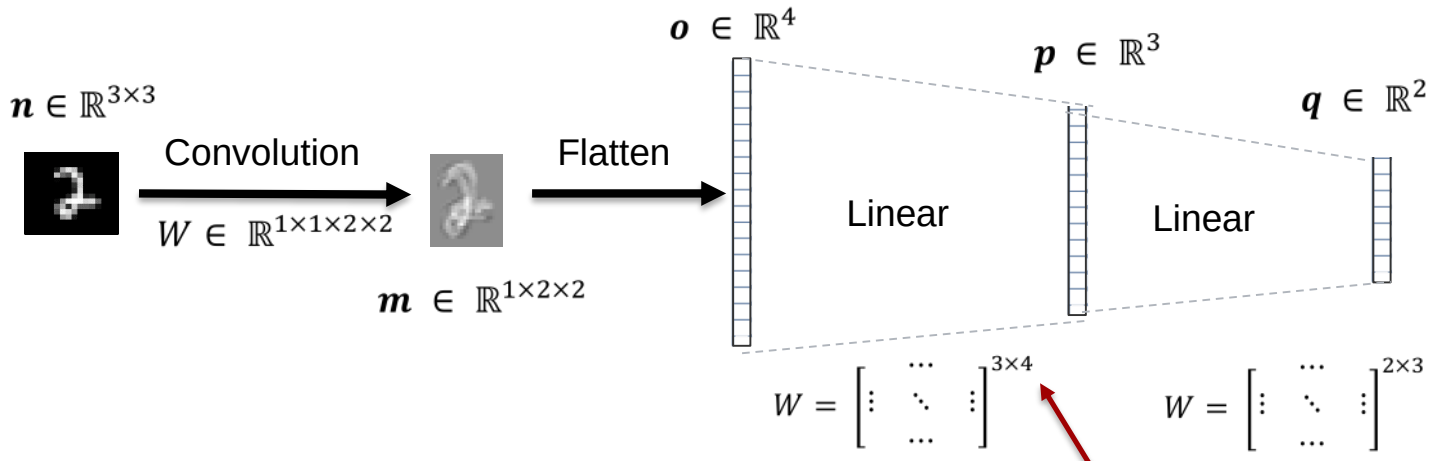


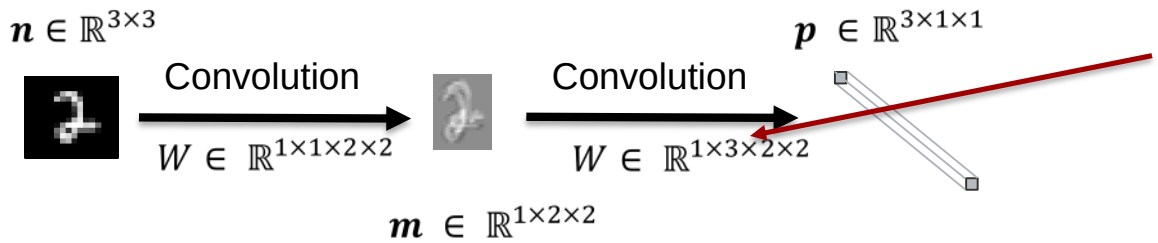


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Everything is a Convolution!



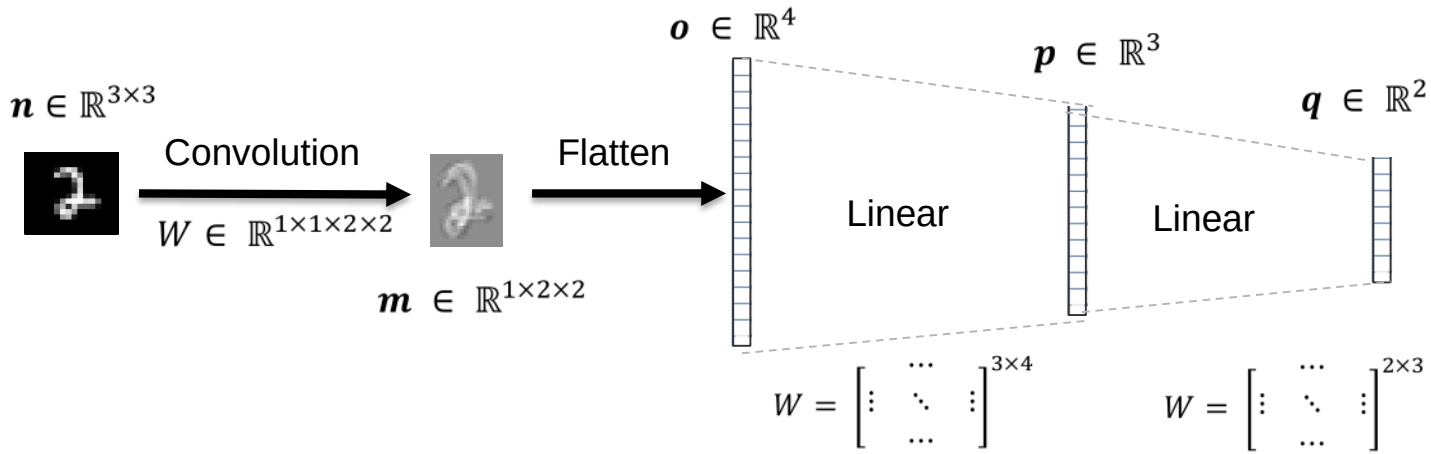


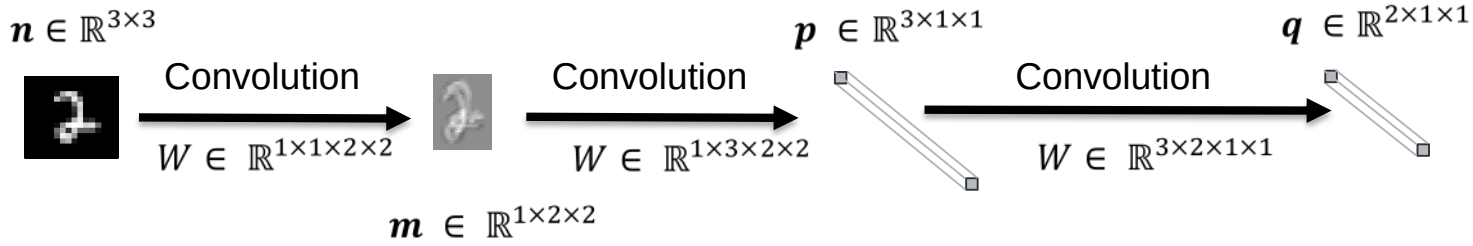
Each output channel (W[:, I, :, :]) corresponds to a row I of the matrix

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Everything is a Convolution!



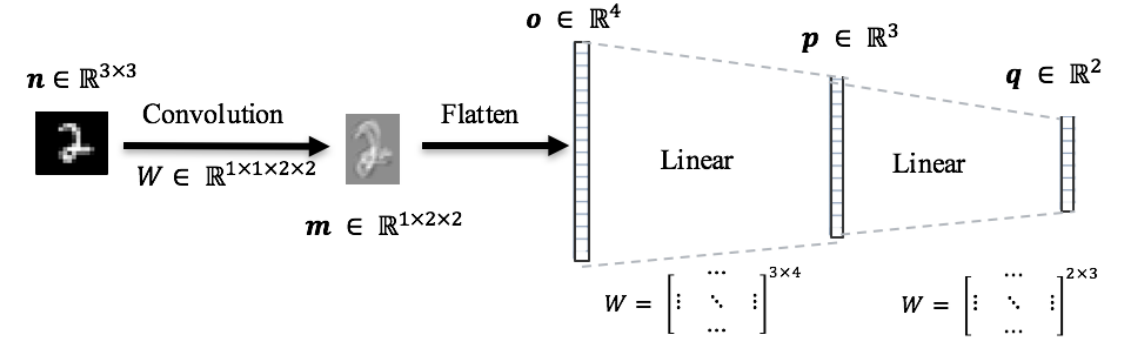


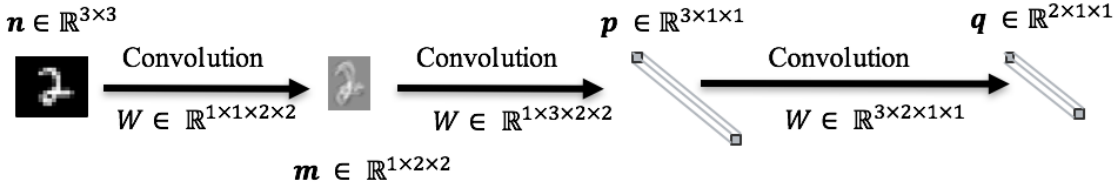
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TL;DR: Converting Linear to Convolution

- Do not *flatten*, instead:
- Do a convolution with height and width of the kernel as the size of the features just before the flatten
- Number of input channels as the number of input channels in the features just before the flatten
- Number of output channels as the number of neurons after the linear layer!





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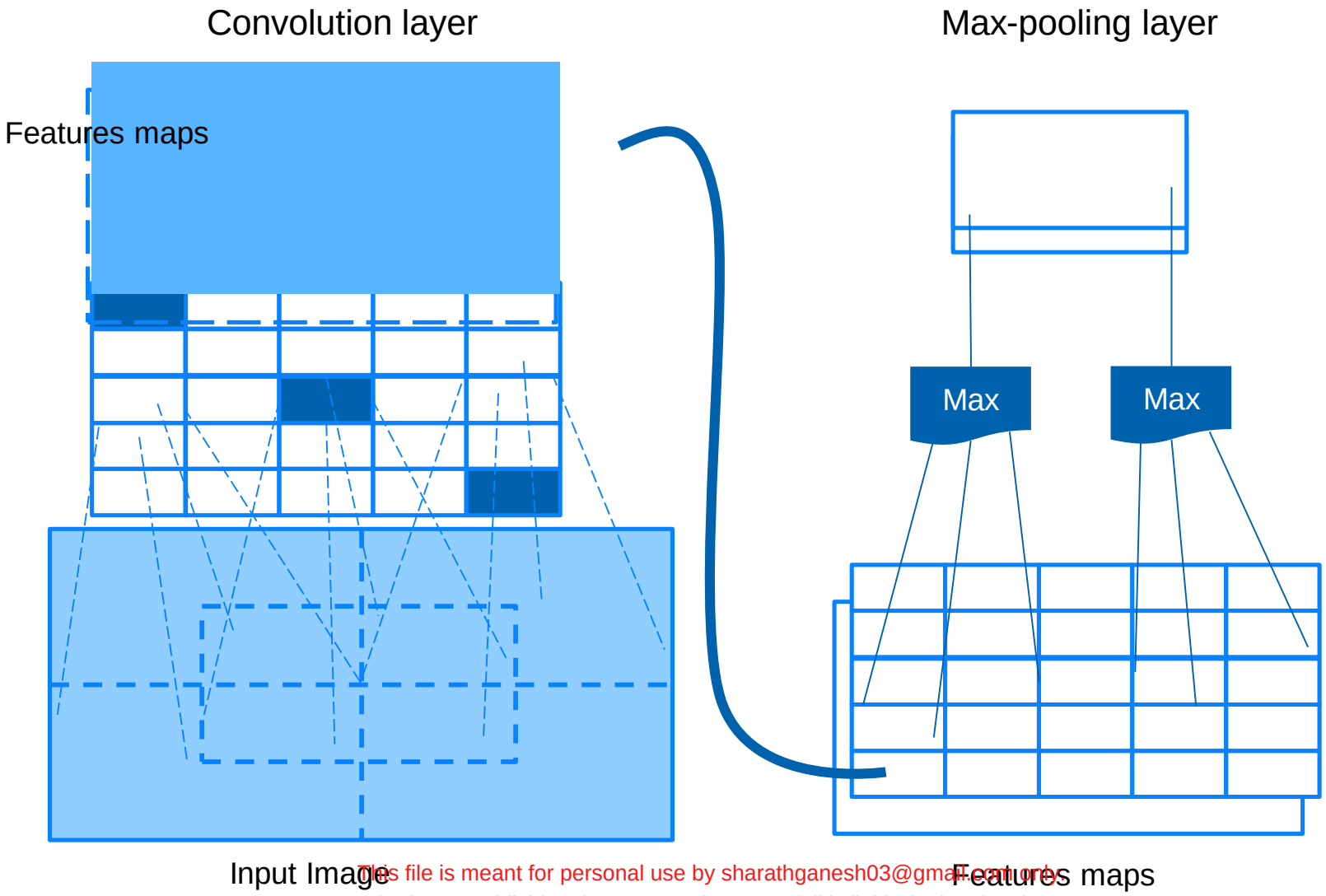
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Max Pooling

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Pooling layer



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Pooling in action

- Pooling doesn't change the depth. It only affects the length and the width of the input.
- It introduces no parameters.

Max Pooling

38	20	77	33
36	49	12	48
26	65	100	40
90	31	43	18

2x2 Pool size

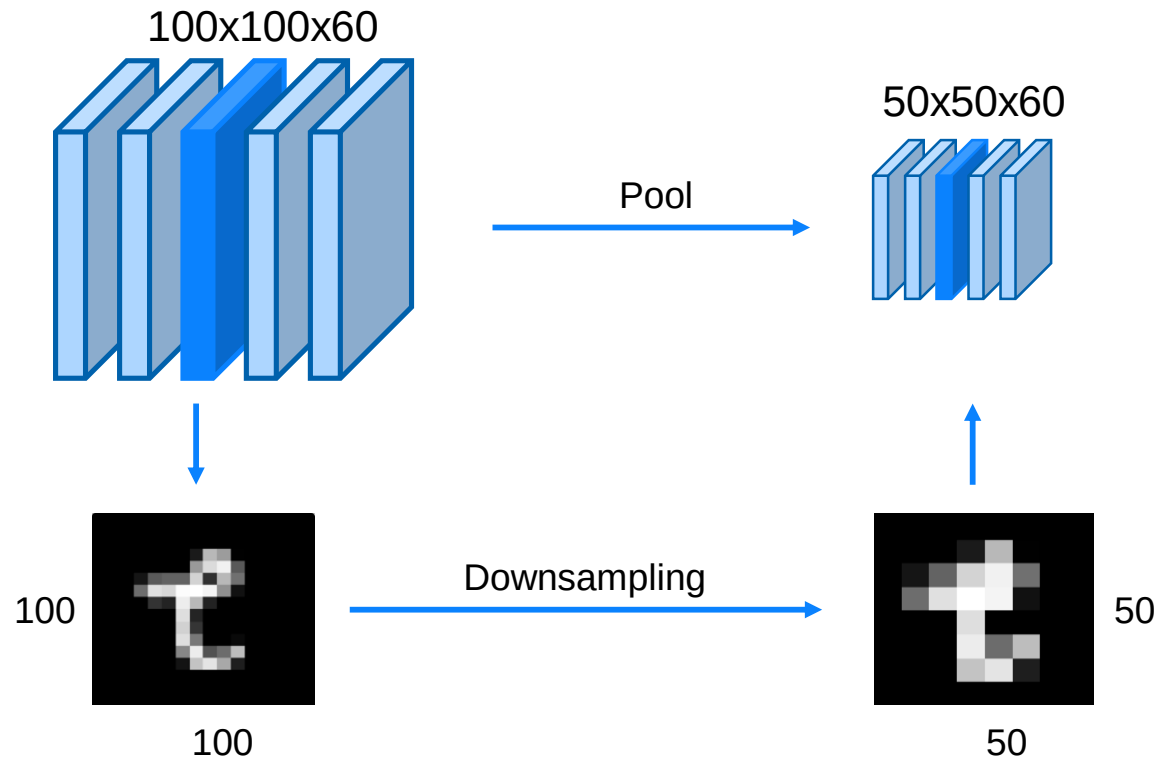
49	77
90	100

Average Pooling

38	20	77	33
36	49	12	48
26	65	100	40
90	31	43	18

2x2 Pool size

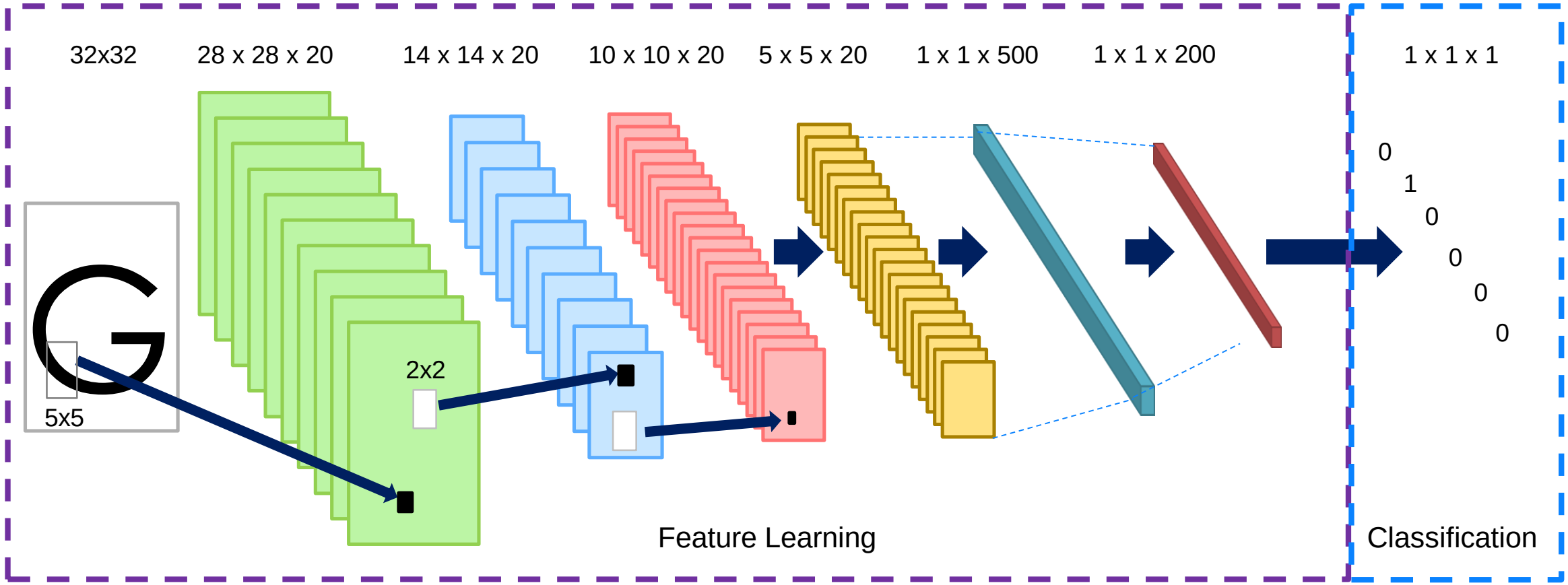
36	43
53	50



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Convolution Network vs. Plain Neural Network

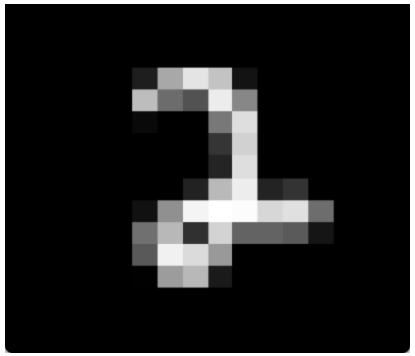


Input Image 1 Map Neurons 1024	CONV Layer Filters = 20 Kernel 5x5	POOL Layer Kernel 2x2 Max Pool	CONV Layer Filters = 20 Kernel 5x5	POOL Layer Kernel 2x2 Max Pool	FC Layer	FC Layer	Output layer Fully Connected

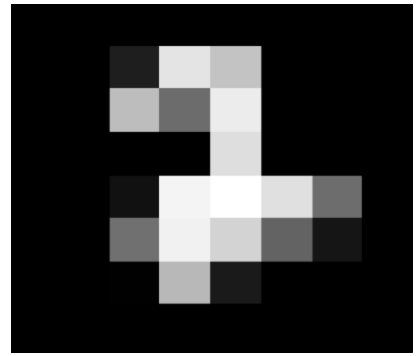
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Pooling (Max Pooling)



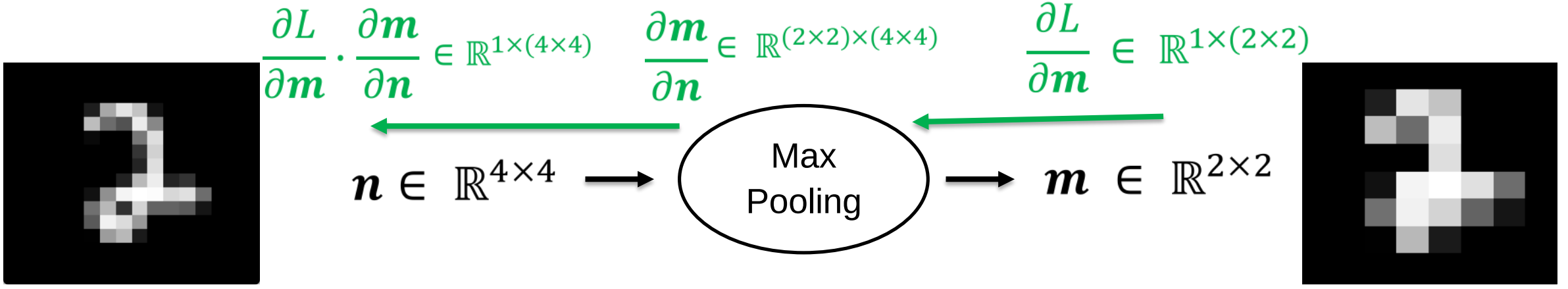
$$n \in \mathbb{R}^{4\times4} \rightarrow \text{Max Pooling} \rightarrow m \in \mathbb{R}^{2\times2}$$



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Pooling (Max Pooling) in Tensorflow




```
n = tf.random_normal((1,4,4,1))
pool = tf.layers.MaxPooling2D([2,2],[2,2],"valid")
nextgrad = tf.ones((1,2,2,1))
with tf.GradientTape(persistent=True) as t:
  t.watch(n)
  m = pool(n)
  m1 = tf.multiply(nextgrad,m)
print(n)
```

```
tf.Tensor(
[[[[ 0.58009243]      [[ 0.53753567]
   [ 1.1895669 ]       [ 0.10521288]
   [ 2.6777048 ]       [-0.43827865]
   [ 0.15597591]]      [-0.7727399 ]]

  [[-0.94100916]      [[-0.42974523]
   [-0.7437242 ]       [-1.669584  ]
   [-1.0578039 ]       [-0.92768306]
   [-0.30651447]]      [-0.19861951]]]],

  shape=(1, 4, 4, 1), dtype=float32)
```

```
print(m)
```

```
tf.Tensor(
[[[[ 1.1895669 ]
   [ 2.6777048 ]]

  [[ 0.53753567]
   [-0.19861951]]]], shape=(1, 2, 2, 1), dtype=float32)
```

```
gradInput = t.gradient(m1,n)
print(gradInput)
```

```
gradInput = t.gradient(m1,n)
print(gradInput)
```

```
tf.Tensor(
[[[[0.]      [[1.]
   [1.]       [0.]
   [1.]       [0.]
   [0.]]      [0.]]

  [[0.]      [[0.]
   [0.]       [0.]
   [0.]       [0.]
   [0.]]      [1.]]]],

shape=(1, 4, 4, 1), dtype=float32)
```

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Other Pooling layers

- No Pooling?

Striving for Simplicity: The All Convolutional Net
Jost Tobias Springenberg, Alexey Dosovitskiy, Thomas Brox, Martin Riedmiller

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Thank you!

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